MIL-S-36885A
10 August 1978
SUPERSEDING
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MILITARY SPECIFICATION

SHAKING MACHINE, LABORATORY, Slide (Rotating Type), 115 volt, 60 Hz, AC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers one type of constant speed slide rotator that is capable of accommodating four 3 x 2 inch or two 4-3/4 x 3-1/8 inch serological ring slides, for use in medical laboratory serological procedures.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Setup.

MILITARY

MIL-L-7298 - Manual, Technical, Commercial Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

UNDERWRITERS' LABORATORIES, INC.

UL-73 - Motor-Operated Appliances.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L.I., NY 11746; 207 East Ohio Street, Chicago IL 60611 and 1655 Scott Boulevard, Santa Clara, CA 95050).

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of UL-73. Proof shall consist of one of the following:

(a) UL label or UL listing of the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the Contracting Officer, upon the latter's request.

(b) Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample.

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This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified.

3.1.2 Compliance with paragraph 3.1 does not excuse compliance with all other requirements of the contract and specifications.

3.2 Materials. Material used shall be of the highest quality throughout. All metal parts used shall be corrosion-resisting.

3.3 Design and construction. The serological slide rotator shall consist of a horizontal rotating platform attached to the moving ends of four rotating type arms that are mounted (i.e. pivot ends) on top of a control box. The control box shall be a housing that essentially contains a motor, gear train, recording digital counter with a reset thumb wheel, timing device, on-off switch, motor starter switch, and power cord. The motor shall drive the gear train, which, in turn, shall concurrently control the following:

(a) The rotation of the arms and platform.
(b) The operation of the timing device (see paragraph 3.3.5).
(c) The operation of the recording digital counter (see paragraph 3.3.6).

The rotation of the platform shall be controlled by the motion of the arms. The rotation of the arms shall result in each point on the platform circumscribing a 3/4 inch diameter circle. The platform and rotating arms shall be so oriented that no part of the platform shall exceed or go past the edges of walls of the control box at any point during the circumscribing of the full revolution of the platform. The platform shall rotate at a constant speed of 180 r.p.m. during a continuous cycle indicated in paragraph 3.3.5. The overall dimensions of the assembled rotator shall be 8-3/4 wide by 8-1/4 deep by 4-1/4 high (inches) with a tolerance of $\pm$ 1/4 on all dimensions. In addition, the weight of the assembled rotator shall be 6 $\pm$ 3/4 pounds (lbs).

3.3.1 Housing (control box). The sides and bottom of the housing shall be fabricated of 1/64 inch minimum steel sheet. The bottom and at least two sides shall have sufficient ports for ventilation and cooling of the internal components. The top surface of the housing shall be fabricated of 1/16 inch minimum aluminum that is anodized. The bottom of the housing shall be provided with rubber feet or pads that are mounted at each corner. The top surface shall have cutouts for mounting the counter and reset wheel, on-off switch, starter switch, and rotating arms for the platform (see paragraph 3.3). The back or one side of the housing shall have a cutout to accommodate the power cord (see paragraph 3.3.8). In addition, the top surface plate shall be removable to allow access to the internal components.

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3.3.2 Platform. The platform shall consist of a bakelist panel enclosed in a frame with a projecting rim. The frame shall be formed from aluminum channel 1/16 inch thick and 1/2 x 1/2 inch wide. The panel shall be fitted and riveted to the horizontal leg of the frame in such a manner as to prevent buckling or warping. The top side shall be covered with a spongy non-slip neoprene pad. The underside of the panel shall have suitable mountings for attachment to the rotating arms (see paragraph 3.3). The vertical leg shall serve as a protective wall to prevent the slides from slipping during rotation. The platform shall be rectangular in shape, and shall be not less than 5 by 6-1/2 inches when measured on the inner side of the frame. The corners of the platform may be rounded.

* 3.3.3 Motor. Shall be a single phase induction motor suitable for operation on 90 to 125 volt, 60 Hz, AC supply lines. The motor shall have a current rating that does not exceed 0.75 amperes at 120 volts. The torque capabilities of the motor shall be such that the operation of the rotating platform results in no significant departure from the no-load motor speed. The motor operation shall be initiated by the starter switch specified in paragraph 3.3.4.

3.3.4 Motor starter switch. Shall be a push-button type switch that starts the motor operation when the on-off switch (see paragraph 3.3.7) is in the "on" position. The service data of pargraph 3.7 shall indicate how long the switch is to be held down until the timing device of paragraph 3.3.5 takes control of the operation of the platform. The top mounting plate of paragraph 3.3.1 shall be permanently marked in a position near the switch to indicate its function. At the end of an operating cycle, the platform shall not rotate until the starter switch initiates a new cycle.

3.3.5 Timing device. The timing device shall control the operating cycle of the rotator. At the end of the operating cycle, an automatic shut-off switch shall cause the rotator to cease operating. When tested in accordance with paragraph 4.4.1 the platform shall have a speed of 180 r.p.m $\pm$ 10 rpm; a continuous operating cycle of 4 minutes $\pm$ 10 seconds and a total of 720 $\pm$ 5 revolutions shall have been completed at the end of the operating cycle. If power to the motor is removed before completion of the operating cycle, the timing device will not be required to advance to the shut-off point. When power to the motor is restored, the platform can resume rotating (i.e. without using the starter switch of paragraph 3.3.4) until the shut-off point is reached to complete the operating cycle.

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3.3.6 Counter. Shall be a recording digital counter that counts and registers the number of rotations of the platform. The counter shall have three numerical wheels that register from "000" to "999". The thumb wheel shall be sufficiently exposed in the top place (see paragraph 3.3.1) to permit manual resetting of the counter to "000" after completion of each cycle. The numerical wheels shall be mounted so that the three significant digits are exposed through a window in a cutout in the top plate.

3.3.7 On-off switch. The on-off switch shall control the power from the line cord (see paragraph 3.3.8) to the starter switch of paragraph 3.3.4. The switch or housing plate (of paragraph 3.3.1) shall be permanently marked to indicate the on and off positions of the switch.

3.3.8 Power cord. A three-conductor power cord with non-detachable plug shall be supplied. Strain relief shall be provided where the cord enters the housing of paragraph 3.3.1. The cord shall be equipped with a three-blade make plug described in ANSI C73.11. The overall exposed length of cord and plug shall not be less than six feet. The interconnection of the power cord to the housing shall provide grounding of metal surfaces for protection of personnel from shock. There shall be a positive indication of continuity of ground when tested in accordance with paragraph 4.4.2.

3.3.9 Operation. When tested in accordance with paragraph 4.4.3, the rotator shall operate in accordance with the following:

(a) The platform continues to rotate after the starter switch button is released.
(b) The platform and counter operate without any type of failure.
(c) The voltage fluctuations do not appreciably affect the rotation of the platform.
(d) No part of the platform exceeds or goes past the edges of the walls of the control box.

There shall be no electric breakdown when the rotator is tested in accordance with paragraph 4.4.4.

3.4 Finish. The steel members of the control box shall have an enamel finish. The manufacturer's commercial colors shall be acceptable.

* 3.5 Identification marking. The rotator shall be legibly and permanently marked with the National Stock Number, contract or purchase order number, name or registered trademark of the manufacturer, model or catalog number and electrical characteristics (amperes, volts and cycles). Marking secured by adhesives shall not be allowed.

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3.6 Workmanship. Workmanship shall be first class throughout and shall be free of defects which detract from performance and/or appearance. There shall be no evidence of improper or incomplete wiring or illegible marking. The rotator shall be free of sharp edges, sharp corners, and dents. All internal electric wiring shall be adequately secure to prevent breaking off at terminations. Any soldering used shall result in no cold solder joints. Any mechanical terminations that are used shall be adequate to prevent breaking off or slipping off.

3.7 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of 15 additional copies shall be the option of the Government.

4. QUALITY ASSURANCE PROVISIONS

★ 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. For examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Tests shall be conducted on the rotator in accordance with MIL-STD-105 and table II.

TABLE II. For tests.

Characteristic	Requirement	Test procedure	Inspection level	AQL
Operating cycle	3.3.5	4.4.1	S-1	0
Continuity of ground	3.3.8	4.4.2	100%	-
Operation	3.3.9	4.4.3	100%	-
Dielectric strength	3.3.9	4.4.4	100%	-

4.3 Examination. The rotator shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects shown in table III. Examination shall not be restricted to the classified possible defects listed in the table.

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TABLE III. Classification of defects.

Categories	Defects
Major	
101	Rotator not complete with all component parts.
102	Components not free of cracks, tears, dents, deformation, or corrosion.
103	Rotator not free of rocking or wobbling when placed on a flat and level surface.
104	Components not free of sharp edges or burrs.
105	Counter reset wheel not capable of being adjusted throughout the complete range.
106	Exposed surface of neoprene pad not free of cuts, tears, or tackiness.
107	Components not capable of being adequately secured to platform or housing.
108	Line cord not adequately secured to housing.
109	Line cord not free of cuts, tears, or abrasion.
110	Blades of line cord plug not properly aligned.
Minor	
201	Identification marking not complete, legible, or permanent.
202	Finish not free of scratches, chipping, or blisters.
203	Components not free of dirt, grease or foreign matter.

4.3.2 Dimensional examination. The rotators shall be examined for defects in dimensions. Any dimension not within the specified limits shall be classified as a defect.

* 4.3.3 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

* 4.4.1 Operating cycle. The platform shall hold either four 3 x 2 inch or two 4-3/4 x 3-1/8 inch serological ring slides. The timing device shall be at the shut-off point, and the power cord shall be connected to an outlet having 115 ± 5 volts, 60 Hz, AC. With the counter set to the "000" position and the on-off switch in the "on" position, the starter switch shall be activated. A stopwatch shall be used to determine the time for the rotating platform to complete the four minute cycle; a stroboscope or other suitable means shall be used to determine the speed of the rotating platform. After the platform rotation stops, the final reading of the counter shall be recorded.

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4.4.2 Continuity of ground. A suitable continuity indicating device (e.g. ohmmeter or battery and buzzer combination) shall be connected between the ground conductor in the plug and the control box.

4.4.3 Operation. The platform shall hold either four 3 x 2 inch or two 4-3/4 x 3-1/8 inch serological ring slides. The timing device shall be at the shut-off point. The power cord shall be connected to a variable voltage supply having a range that includes 90 to 125 volts, 60 Hz, AC. With the on-off switch in the "on" position, the starter switch shall be activated to rotate the platform. While the platform is rotating, the voltage shall be slowly varied to cover the 90 to 125 voltage range. The platform and counter shall be checked for proper operation.

* 4.4.4 Dielectric strength. With the on-off switch in the "off" position, the rotator shall be subjected to a gradually increasing 60 Hz potential to 1000 volts AC, that is applied between the current carrying parts and the metal portions of the control box. The maximum value of applied voltage shall be maintained for a period of one minute.

* 5. PACKING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One shaking machine constitutes a unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging. Each shaking machine shall be packaged in a box conforming to PPP-B-636, class domestic. The unit shall be cushioned within the box by use of fiberboard pads or by other suitable cushioning medium. Closure shall be as specified in the appendix of the box specification.

5.1.3 Level C. Each shaking machine shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Eight unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

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5.2.2 Level B. Eight unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carriers' rules and regulations.

5.3 Marking. Each unit package and shipping container shall be marked as specified in MIL-STD-129.

* 6. NOTES

6.1 Intended use. The item covered by this specification is a serological slide rotator suitable for use in medical laboratory procedures.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

6.4 This specification covers the following item shown in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-135-9644	SHAKING MACHINE, LABORATORY, Slide (Rotating Type), 115 volt, 60 Hz, AC

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6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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